

DATA266 Lab 1 — Team 33 Report (Tejas's version)

- **Repository:** https://github.com/SawantTej2912/DATA266_Lab
- **Members:** Tejas Nandkishor Sawant (*tejas/), Shriram Dundigalla (*shriram_dundigalla/)
- **Kaggle team:** PairProgramming_Team_33

All numbers in this report are taken from the repository file cited with each table. Values marked "(derived)" are simple arithmetic on cited values. Confidence intervals are written as [low, high].

Ownership statement

Each member designed, trained and evaluated his own models for all three tasks, and no model was shared or trained jointly. Before training we agreed on a shared measurement protocol (TEAM_PROTOCOL.md, drafted by Shriram and agreed by both). Tejas built a character-level GPT (6 layers, 6 heads, 384-d, 256-character context) for Task 1; an n-gram bag, a Transformer encoder and a BiGRU with attention for Task 2; and a ResNet-9 CycleGAN (run i270k) for Task 3. Shriram built a character-level GPT (6 layers, 8 heads, 256-d, 128-character context) for Task 1; a bag-of-embeddings MLP, a BiLSTM and a TextCNN for Task 2; and a U-Net CycleGAN (run 1, epoch 61) for Task 3. He also wrote the shared tooling in scripts/. Both members submitted to Kaggle as one team, PairProgramming_Team_33. Tejas's Task 3 audit was scored by two raters (Rater 1: Tejas Nandkishor Sawant; Rater 2: Shriram Dundigalla); the audit of Shriram's outputs is pending.

Summary of results

Task	Tejas: best model, headline metric	Shriram: best model, headline metric	Team best
1. Character-level GPT	GPT 6L/6H/384d: val perplexity 1.7482, 0.8059 bits/char	GPT 6L/8H/256d: val perplexity 2.1045, 1.0735 bits/char	Tejas (perplexity 1.7482)
2. Yelp sentiment	BiGRU + attention: test accuracy 0.9536, MCC 0.9073	TextCNN: test accuracy 0.9344, MCC 0.8688	Tejas (accuracy 0.9536)
3. CycleGAN	ResNet-9 (i270k): submitted FID 98.70, Kaggle score −49.55	U-Net (run 1): submitted FID 101.44, Kaggle score −50.93	Tejas (FID 98.70; team rank 20)

Tejas's models lead on every headline metric, although training data, context length and training budget also differ between members (values from each member's metrics_report.csv; Shriram's Kaggle score from his Task 3 notebook).

Comparability notes

Some choices differ from TEAM_PROTOCOL.md. They affect how the side-by-side tables should be read.

Task	Protocol	What was done	Source
1	100K / 10K windows, shared metrics.py, 400-character probe at T in {0, 0.5, 0.8, 1.0, 1.2}	Shriram followed the protocol. Tejas used 100K / 10K stories (about 5 times the text), his own metric code, a 300-character probe with greedy and T = 0.8 only, and distinct-n pooled over 45 samples	task1_llm/tejas/src/task1_llm.ipynb (CFG cell), task1_llm/shriram_dundigalla/src/config.yaml
2	Balanced 100K training subsample, 2,000 bootstrap replicates, shared slices	Shriram followed the protocol. Tejas used the full 560K training split (5% held out for validation), 1,000 bootstrap replicates and his own slices (contrast, negation, length quartile, truncated)	task2_sentiment/tejas/metrics_report.csv, task2_sentiment/tejas/src/task2_sentiment.ipynb, task2_sentiment/shriram_dundigalla/src/config.yaml
3	LPIPS over random generated pairs (diversity)	Both members report LPIPS between each input and its translation, so the two members are comparable with each other but not with the protocol definition	task3_gan/tejas/evaluate_local.py, task3_gan/shriram_dundigalla/results.md
All	Same hardware not required	Tejas used an RTX 4090 for all tasks. Shriram used an Apple M4 (MPS) for Tasks 1 and 2 and a Colab A100 for Task 3. Time, throughput and memory are not directly comparable across members	each metrics_report.csv

Task 1: GPT from scratch (TinyStories, character level)

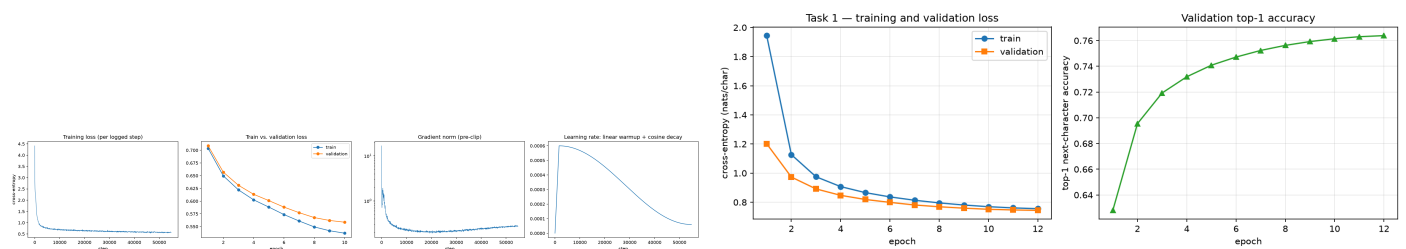
Comparison table

	Tejas (t1_gpt_tejas)	Shriram (gpt_char_tinystories_v1)
Architecture	Decoder-only GPT, 6 pre-LN blocks, 6 heads, d_model 384, FFN 4x GELU, learned token and position embeddings, LM head tied to token embedding	Decoder-only GPT, 6 pre-LN blocks, 8 heads, d_model 256, d_ff 1024, learned token and position embeddings, untied LM head
Context length	256 characters	128 characters
Vocabulary	77 symbols (characters seen fewer than 50 times mapped to one symbol)	98 symbols (index 0 = <unk>)
Data	100,000 train / 10,000 val stories, seed 1446	100,000 train / 10,000 val windows from 20,000 stories (split by story), seed 9015
Dropout	0.1	0.1
Optimiser	AdamW (0.9, 0.95), weight decay 0.1 on weight matrices, clip 1.0	AdamW (0.9, 0.95), weight decay 0.1 on matmuls only, clip 1.0
LR schedule	peak 6e-4, 3% linear warm-up, cosine to 6e-5	peak 3e-4, 3% linear warm-up (281 of 9,372 steps), cosine to 0.1x
Batch / epochs	64 x 256 characters, 10 epochs (54,850 steps), fp16 AMP	128 x 128 characters, 12 epochs (9,372 steps), fp32
Parameters	10,775,424	4,822,016
Train CE (nats/char)	0.5370	0.7089
Val CE (nats/char)	0.5586	0.7441

	Tejas (t1_gpt_tejas)	Shriram (gpt_char_tinystories_v1)
Val perplexity	1.7482	2.1045 (train 2.0317)
Val bits/char	0.8059	1.0735 (train 1.0227)
Generalization gap (val CE minus train CE)	0.0216	0.0352
Top-1 next-character accuracy (val)	0.8212 (train 0.8267)	0.7639 (train 0.7728)
Best epoch	10 of 10 (val still falling)	12 of 12 (val still falling)
Distinct-1 / 2 / 3 (word level, T = 0.8)	0.2727 / 0.6934 / 0.8901, pooled over 45 continuations	single draw 0.7073 / 0.9877 / 1.0000; mean $\pm$ SD over 15 draws 0.714 ± 0.046 / 0.957 ± 0.030 / 0.991 ± 0.014
Repeated 4-gram rate (T = 0.8)	0.0036	single draw 0.0000; 15 draws 0.0017 ± 0.0044
Greedy: distinct-2 / repeated 4-gram	0.6278 / 0.0804	0.690 ± 0.311 / 0.235 ± 0.319 (15 draws)
Best decoding strategy	not evaluated (greedy and T = 0.8 only)	repetition penalty 1.15 (repeated 4-gram 0.000 ± 0.000)
Gradient norm mean / max	0.280 / 16.77 (max at step 0)	0.701 / 11.13 (max at step 0)
NaN / non-finite	18 fp16 overflow steps skipped by GradScaler, 0 NaN losses	0
Loss spikes	0	0
Train throughput	357,646 tokens/s	24,990 tokens/s
Generation throughput	256.7 tokens/s (no KV cache)	371.8 tokens/s
Peak memory	3,324 MB (CUDA)	80.0 MB (MPS allocator)
Training time	45.4 min compute (47.4 min wall)	6,144 s (102.4 min)
Hardware	RTX 4090 + Ryzen 9 7950X	Apple M4 (MPS)

Sources: task1_llm/tejas/metrics_report.csv, task1_llm/tejas/outputs/epoch_history.csv, task1_llm/tejas/results.md; task1_llm/shriram_dundigalla/metrics_report.csv, task1_llm/shriram_dundigalla/results.md (15-draw sweep and decoding tables), task1_llm/shriram_dundigalla/src/config.yaml.

Shriram's generation throughput is taken from his metrics_report.csv; his results.md lists 387 tokens/s.



Left: Tejas, training loss, train and val loss per epoch, gradient norm and LR. Right: Shriram, train and val loss curves.

Joint analysis

Strengths. Tejas's model has the lower validation CE (0.5586 vs. 0.7441) and the higher top-1 accuracy (0.821 vs. 0.764). Both runs were stable, with 0 loss spikes and the maximum gradient norm at step 0. Tejas's 18 non-finite steps are fp16 overflows (0.03% of steps), not divergence. Both generalization gaps are small (0.0216 and 0.0352). Sampling at $T = 0.8$ removes most of the greedy repetition, and Shriram showed that a repetition penalty of 1.15 removes repeated 4-grams entirely.

Weaknesses. Greedy decoding loops for both models (repeated 4-gram rate 0.080 for Tejas, 0.235 ± 0.319 for Shriram). Both failure analyses trace story resets to the context window (128 characters for Shriram; 256 characters, shorter than the 300-character generations, for Tejas). Both models still underfit: validation loss was still falling in the last epoch (Tejas by 0.0035 in epoch 10, Shriram by 0.0032 in epoch 12, derived from the epoch logs).

Limitations. The two models differ in size (10.78M vs. 4.82M parameters), context (256 vs. 128 characters), training text (about 5 times more for Tejas) and vocabulary (77 vs. 98 symbols), so the perplexity gap cannot be assigned to a single factor. Absolute distinct-n values are not comparable, because Tejas pools n-grams over 45 continuations and Shriram averages over 15 draws. Throughput and memory differences mostly reflect the hardware (RTX 4090 vs. Apple M4).

Next steps. Train both models for more epochs or with more capacity. Try a longer context or subword tokens, add a KV cache, use a repetition penalty for both models, and rerun both on matched data.

Evidence

	Tejas	Shriram
Loss curves	loss_curves.png	loss_curves_...png , stability_...png
Raw log	t1_gpt_tejas_20261001-144225.jsonl	task1_gpt_char_tinystories_v1_20260928T204206Z.log
Generations	generations.json , diversity_samples.json , failure_candidates.json	generations_...txt , generation_sweep_...csv , decoding_comparison_...csv
Checkpoint	best.pt (epoch 10)	gpt_char_tinystories_v1_20260928T204206Z.pt
Manifest	manifest_t1_gpt_tejas.json	shriram_dundigalla_manifest.md

Task 2: Yelp Polarity sentiment classification

All six models are scored on the full official test set of 38,000 reviews (19,000 per class) at threshold 0.5. No pretrained embeddings are used.

Comparison table

	T-Base n-gram bag	T-Exp1 Transformer	T-Exp2 BiGRU + attention	S-Base bag of embeddings	S-Exp1 BiLSTM	S-Exp2 TextCNN
Member	Tejas	Tejas	Tejas	Shriram	Shriram	Shriram
Architecture	mean of unigram and hashed-bigram embeddings, linear output (200K buckets)	2 layers, 4 heads, d 128, FFN 256, learned positions, pre-LN, masked mean pool	BiGRU hidden 128 with attention pooling	mean-pooled embeddings, MLP [256], 1 logit	BiLSTM hidden 192, [mean; max] pool, 1 logit	Conv1d widths 2/3/4/5 with 128 filters, global max pool, 1 logit
Embedding dim	64	128	128	128	200	160
Dropout / LR / batch	0.2 / 2e-3 / 256	0.1 / 5e-4 / 128	0.3 / 1e-3 / 128	0.3 / 1e-3 / 256	0.4 / 1e-3 / 128	0.5 / 1.5e-3 / 256
Epochs (max / run)	5 / 5	5 / 5	5 / 5	6 / 3	5 / 5	6 / 4
Optimiser	Adam, clip 5.0, early stopping (patience 2)	same	same	AdamW wd 1e-4, clip 1.0, early stopping (patience 2)	same	same
Preprocessing	lemmatize, keep negations, vocab 45,818 (min_freq 5), max_len 188, head and tail truncation	same	same	Porter stem, keep negations, vocab 30,000 (min_freq 2), max_len 250	same	same
Training data	full 560K (5% val)	same	same	balanced 100K + 10K val	same	same
Accuracy [95% CI]	0.9327 [0.9301, 0.9351]	0.9365 [0.9341, 0.9388]	0.9536 [0.9514, 0.9557]	0.9245 [0.9218, 0.9272]	0.9334 [0.9309, 0.9358]	0.9344 [0.9319, 0.9368]
Macro P / R / F1	0.9327 / 0.9327 / 0.9327	0.9365 / 0.9365 / 0.9365	0.9536 / 0.9536 / 0.9536	0.9247 / 0.9245 / 0.9245	0.9337 / 0.9334 / 0.9334	0.9344 / 0.9344 / 0.9344
Macro-F1 95% CI	[0.9301, 0.9351]	[0.9341, 0.9388]	[0.9514, 0.9557]	[0.9218, 0.9271]	[0.9309, 0.9358]	[0.9319, 0.9368]
Micro P / R / F1	0.9327	0.9365	0.9536	0.9245	0.9334	0.9344
Weighted P / R / F1	0.9327 / 0.9327 / 0.9327	0.9365 / 0.9365 / 0.9365	0.9536 / 0.9536 / 0.9536	0.9247 / 0.9245 / 0.9245	0.9337 / 0.9334 / 0.9334	0.9344 / 0.9344 / 0.9344
Confusion TN / FP / FN / TP	17,786 / 1,214 / 1,345 / 17,655	17,902 / 1,098 / 1,316 / 17,684	18,143 / 857 / 905 / 18,095	17,756 / 1,244 / 1,625 / 17,375	17,498 / 1,502 / 1,028 / 17,972	17,759 / 1,241 / 1,251 / 17,749
ROC-AUC	0.9794	0.9843	0.9906	0.9751	0.9830	0.9829
PR-AUC	0.9788	0.9847	0.9909	0.9748	0.9836	0.9835

	T-Base n-gram bag	T-Exp1 Transformer	T-Exp2 BiGRU + attention	S-Base bag of embeddings	S-Exp1 BiLSTM	S-Exp2 TextCNN
MCC [95% CI]	0.8653 [0.8603, 0.8703]	0.8730 [0.8683, 0.8777]	0.9073 [0.9028, 0.9114]	0.8492 [0.8438, 0.8545]	0.8671 [0.8622, 0.8719]	0.8688 [0.8638, 0.8737]
Brier	0.0507	0.0477	0.0356	0.0572	0.0500	0.0487
ECE, predicted-class confidence (15 bins)	0.0072	0.0121	0.0120	0.0044	0.0165	0.0061
ECE, positive class	not computed	not computed	not computed	0.0075	0.0203	0.0064
McNemar vs. own baseline (b / c, χ^2 , p)	reference	766 / 911, 12.36, 4.4e-4	561 / 1,358, 330.18, 8.8e-74	reference	976 / 1,315, 49.87, 1.6e-12	932 / 1,309, 63.09, 2.0e-15
Best val macro-F1	0.9312	0.9348	0.9527	0.9214	0.9330	0.9351
Parameters	15,732,417	6,154,113	6,063,362	3,873,281	6,605,953	5,087,745
Training time	0.55 min	2.97 min	11.99 min	25.8 s	2,318.6 s	211.1 s
Examples / s	80,405	14,950	3,698	11,610	216	1,895
Peak memory	355 MB (CUDA)	947 MB (CUDA)	440 MB (CUDA)	60 MB (MPS)	102 MB (MPS)	79 MB (MPS)
Hardware	RTX 4090	RTX 4090	RTX 4090	Apple M4	Apple M4	Apple M4
Bootstrap replicates	1,000	1,000	1,000	2,000	2,000	2,000

McNemar b = baseline right and experimental model wrong; c = baseline wrong and experimental model right. Shriram also reports BiLSTM vs. TextCNN: 931 / 969, p = 0.396 (task2_sentiment/shriram_dundigalla/results.md).

Sources: task2_sentiment/tejas/metrics_report.csv, task2_sentiment/tejas/outputs/table_*.csv, task2_sentiment/tejas/results.md; task2_sentiment/shriram_dundigalla/metrics_report.csv, task2_sentiment/shriram_dundigalla/outputs/mcnemar_tests.csv, task2_sentiment/shriram_dundigalla/src/config.yaml, task2_sentiment/shriram_dundigalla/results.md.

Weakest data slice per model

Model	Weakest slice (n)	Macro-F1	Error rate	Note
T-Base n-gram bag	truncated = True (1,843)	0.9111	0.0787	also lowest macro-F1
T-Exp1 Transformer	truncated = True (1,843)	0.9219	0.0695	also lowest macro-F1
T-Exp2 BiGRU	truncated = True (1,843)	0.9294	0.0624	also lowest macro-F1
S-Base BoW	exclamations = no_exclaim (20,364)	0.9120	0.0854	also lowest macro-F1

Model	Weakest slice (n)	Macro-F1	Error rate	Note
S-Exp1 BiLSTM	exclamations = no_exclaim (20,364)	0.9206	0.0776	lowest macro-F1 is no_negation (0.9185)
S-Exp2 TextCNN	exclamations = no_exclaim (20,364)	0.9229	0.0750	lowest macro-F1 is no_negation (0.9207)

Sources: task2_sentiment/tejas/outputs/table_slices.csv, task2_sentiment/shriram_dundigalla/outputs/slice_metrics.csv. The two members use different slice definitions, so slices are compared within a member only. For Tejas's BiGRU the truncated-slice error rate (0.0624) is about 1.3 times its overall error rate (0.0464, derived).

Joint analysis

Model families. Tejas's n-gram bag beats Shriram's bag of embeddings (0.9327 vs. 0.9245); the hashed bigrams are a likely reason, but this is confounded with about 5 times more training data. Among order-aware models the BiGRU (0.9536) is ahead of the Transformer (0.9365), the TextCNN (0.9344) and the BiLSTM (0.9334).

Within each member. Both of Tejas's experimental models beat his baseline significantly, the BiGRU by a large margin ($p = 8.8e-74$) and the Transformer by a small one ($p = 4.4e-4$). Shriram's BiLSTM and TextCNN are statistically tied ($p = 0.396$), and both beat his baseline by about 1 point. Across both members, order-aware models add 1 to 2 points over a bag model that already reaches 92 to 93%.

Cost. The BiGRU is Tejas's slowest model (3,698 vs. 80,405 examples/s for the n-gram bag) and gains 2.1 accuracy points. Shriram's TextCNN matches his BiLSTM at 11 times the speed (211 s vs. 2,319 s). Parameter counts are dominated by embedding tables for both members. Cost is not comparable across members because the hardware differs.

Calibration. All models are well calibrated (confidence ECE at most 0.0165). For both members the bag baseline has the lowest ECE (0.0072 and 0.0044), so the most accurate model is not the best calibrated.

Slices. Truncated reviews (more than 188 tokens) are the weakest slice for all of Tejas's models. Reviews without exclamation marks are the weakest slice for all of Shriram's models. Both members found that negation is not where errors concentrate once negation words are kept: Tejas's BiGRU has only 0.3 points more error on negation reviews, and Shriram's models score higher macro-F1 on them.

Shared errors. The two independent 20-error reviews flagged some of the same test reviews as probable label noise ("Wow love the place...", "my husband had an omelette...") and the same edited review ("EDIT: They really did change the service..."). This supports a label-noise floor under both error rates.

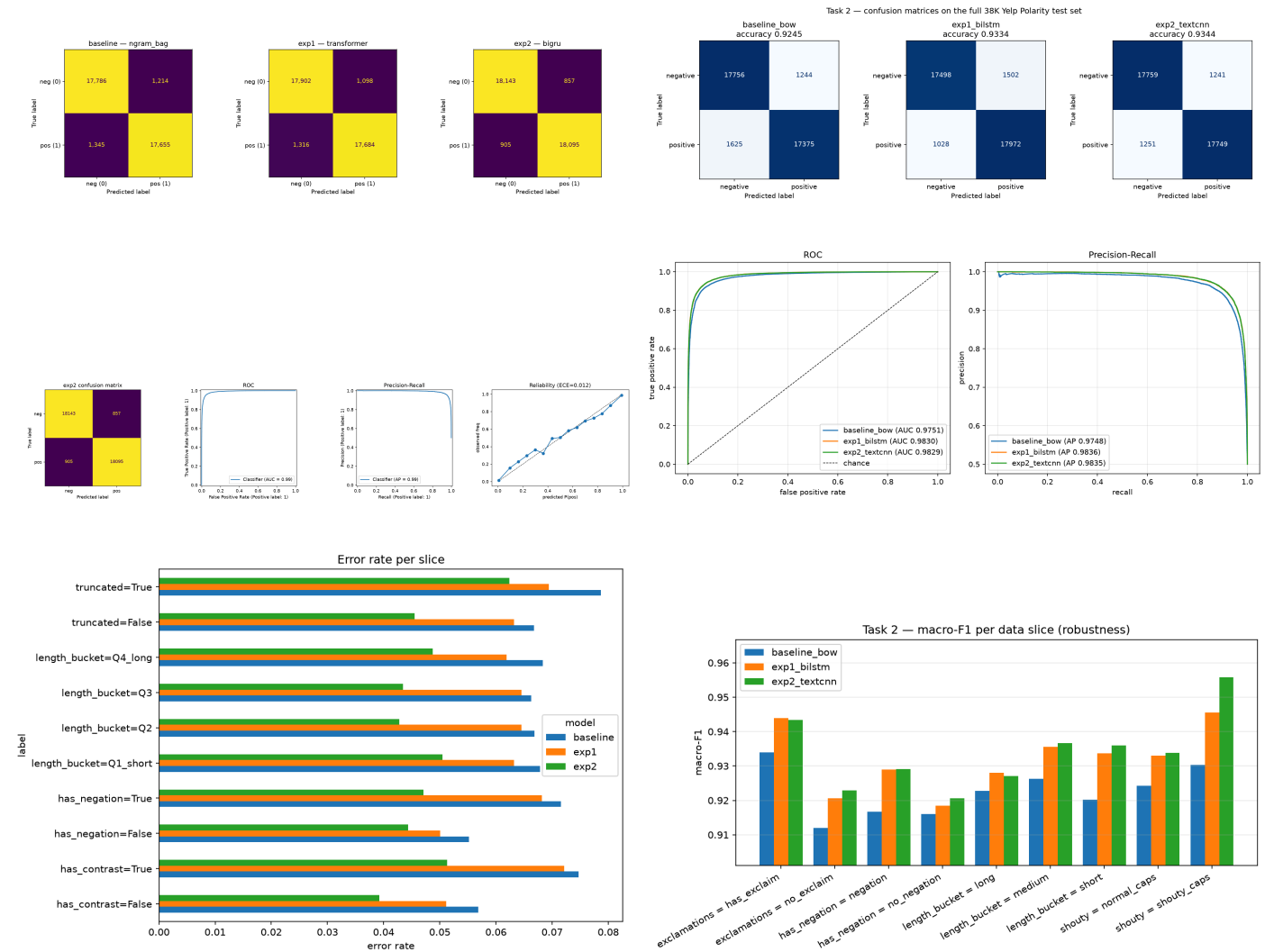
Strengths. Tejas's BiGRU with attention is the best model on every headline metric (accuracy 0.9536, MCC 0.9073, ROC-AUC 0.9906, Brier 0.0356), and its accuracy CI does not overlap any other model's. Every experimental model beats its member's baseline significantly. Shriram's TextCNN matches his BiLSTM at 11 times the speed.

Weaknesses. Long reviews are the BiGRU's main weakness: its error rate on the truncated slice is 0.0624, against 0.0464 overall. Shriram's models are weakest on reviews without exclamation marks (TextCNN error rate 0.0750 on that slice). Shriram's BiLSTM has the highest ECE (0.0165), which matches its rising

validation loss. Part of the remaining error is sarcasm, mixed reviews and label noise, which neither member expects modelling to fix.

Limitations. One seed per model. Tejas trained on all 560K reviews while Shriram used a balanced 100K subsample, so part of the BiGRU's lead may come from data size; a rerun on matched data is needed before calling it an architectural win. Slice definitions, truncation lengths (188 vs. 250 tokens) and bootstrap counts (1,000 vs. 2,000) also differ across members.

Next steps. Retrain each member's best architecture on the same data, run a seed sweep (scripts/seed_sweep_task2.py), test Tejas's max_len 188 to 400 change, test Shriram's position-aware pooling and negation-scope marking, and apply temperature scaling to the BiLSTM.



Left column: Tejas (confusion matrices, BiGRU ROC/PR/reliability, slice error rates). Right column: Shriram (confusion matrices, ROC/PR curves, slice macro-F1).

Evidence

	Tejas	Shriram
Confusion matrices	confusion_matrices_all.png	confusion_matrices.png

	Tejas	Shriram
ROC / PR / reliability	baseline_eval_plots.png , exp1_eval_plots.png , exp2_eval_plots.png	roc_pr_curves.png , calibration.png , bootstrap_ci_s.png
Slices	slice_error_rates.png , table_slices.csv	slice_macro_f1.png , slice_metrics.csv
Training curves	model_comparison.png	training_curves.png
Raw logs	baseline , exp1 , exp2	task2_20260929T033024Z.log
Checkpoints	checkpoints/{baseline,exp1,exp2}_best.pt	checkpoints/{baseline_bow,exp1_bilstm,exp2_textcnn}.pt

Task 3: CycleGAN Monet ↔ Photo

All FID and MiFID values come from the course evaluation script (task3_gan/Part3_Evaluation_Script.ipynb, reproduced in each member's evaluate_local.py): ImageNet normalisation, Resize(299) and CenterCrop(299), $n = 300$ per side, and the submitted value is the mean of both directions. A2B is Monet to Photo and B2A is Photo to Monet.

Comparison table

	Tejas: ResNet-9, run i270k, EMA snapshot epoch 280	Shriram: U-Net, run 1, epoch 61 (ckpt_epoch060.pth)
Generator	ResNet-9 (c7s1-64, d128, d256, 9 residual blocks, 2 resize-convolution upsampling layers, c7s1-3, tanh), InstanceNorm, reflection padding	U-Net depth 8, ngf 64: 8 stride-2 encoder blocks, 8 transposed-conv decoder blocks with skip connections, InstanceNorm, dropout 0.5 in the first 3 decoder blocks, tanh
Discriminator	70x70 PatchGAN, ndf 64, InstanceNorm, LeakyReLU 0.2	70x70 PatchGAN, ndf 64 (C64-C128-C256-C512), 30x30 output map
Parameters	G 11,378,179 each; D 2,764,737 each; 28,285,832 total	G 54,404,099 each; D 2,763,841 each; 114.34M total
Losses	LSGAN + cycle L1 (λ 10) + identity L1 (λ 5 for Monet to Photo; 5 down to 1.5 for Photo to Monet during the decay phase)	LSGAN + cycle L1 (λ 10) + identity L1 (λ 5)
Learning rates	Adam (0.5, 0.999); G 2e-4, D 7.5e-5	Adam (0.5, 0.999); 2e-4 for G and D
Schedule	constant until epoch 135 (45%), then linear decay to 0	constant for 50 epochs, then linear decay to 0 over 50
Iterations	900 per epoch x 300 epochs = 270,000 (selected snapshot at 252,000, derived)	1,000 per epoch x 100 epochs = 100,000 (selected checkpoint at 61,000, derived)
Batch / image pool	1 / 50	1 / 50
Augmentation / regularisation	resize 286, random crop 256, horizontal flip; DiffAugment (colour, translation, cutout) on all D inputs; EMA of G weights (0.999)	resize 286, random crop 256, horizontal flip; N(0, 0.02) init; AMP

	Tejas: ResNet-9, run i270k, EMA snapshot epoch 280	Shriram: U-Net, run 1, epoch 61 (ckpt_epoch060.pth)
Model selection	EMA snapshots every 10 epochs scored with the course FID	checkpoints at epochs 51 to 100 scored with an earlier A2B-only FID; epoch 61 best
Seed / hardware	1446 / RTX 4090	42 / A100 (Colab)
FID A2B / B2A	100.92 / 96.49	103.92 / 98.95
FID submitted (mean)	98.70	101.44
MiFID A2B / B2A / mean	0.4166 / 0.4045 / 0.4106	0.4241 / 0.4104 / 0.4172
KID A2B / B2A	0.0181 ± 0.0022 / 0.0066 ± 0.0015	0.0191 ± 0.0033 / 0.0082 ± 0.0010
Precision / recall A2B	0.747 / 0.483	0.683 / 0.443
Precision / recall B2A	0.543 / 0.720	0.500 / 0.730
Density / coverage A2B	1.078 / 0.907	0.863 / 0.790
Density / coverage B2A	0.451 / 0.710	0.385 / 0.693
Cycle L1 (A→B→A / B→A→B)	0.0785 / 0.0849	0.0471 / 0.0493
LPIS, input vs. translation, A2B / B2A	0.322 / 0.364	0.181 / 0.250
Content cosine A2B / B2A	0.812 / 0.796 (Inception-v3)	0.8259 / 0.7253 (VGG16)
Final cycle / identity loss	0.146 / 0.140 (epoch 300)	0.0935 / 0.0486 (epoch 100)
Final G / D loss	G 2.870; D_A 0.144, D_B 0.184 (epoch 300)	G 2.319; D 0.261 (epoch 100)
NaN count	0	0 (all 3 runs)
G gradient norm mean / max	29.5 / 1,017.9 at epoch 3 (D 17.5 / 72.5)	27.0 / 112.5
Training time	507.0 min (533.2 min wall)	2.77 h (99.6 s per epoch)
Train images / s	17.8	10.04
Inference images / s A2B / B2A	270.9 / 360.8	not measured
Peak memory	3,386 MB (training)	1,882 MB (inference only; training peak not logged)
Kaggle score	−49.55	−50.93
Kaggle rank	team rank 20 (team best score −49.55)	team rank 20 (team best score −49.55)

	Tejas: ResNet-9, run i270k, EMA snapshot epoch 280	Shriram: U-Net, run 1, epoch 61 (ckpt_epoch060.pth)
Human audit (30 samples, 2 raters)	style 4.33 (A2B 4.10 / B2A 4.57), quadratic κ 0.84, 73% agreement; content 4.70 (4.60 / 4.80), κ 1.00, 100%; artifacts in 55% (53% / 57%), κ 0.66, 83%	pending at time of writing (ratings not yet recorded)

Sources: task3_gan/tejas/metrics_report.csv, task3_gan/tejas/results.md, task3_gan/tejas/submission.csv, task3_gan/tejas/outputs/{checkpoint_selection,audit_results}.csv, reproducibility/manifests/task3_gan/tejas/manifest_t3_cyclegan_tejas.json; task3_gan/shriram_dundigalla/{metrics_report,full_metrics_report,submission}.csv, task3_gan/shriram_dundigalla/results.md, and printed outputs of task3_gan/shriram_dundigalla/src/task3_cyclegan_unet.ipynb (per-direction cycle L1, content cosine and KID spread, training seconds, Kaggle score).

Shriram's content cosine uses VGG16 features and is not directly comparable to Tejas's Inception-based value. Shriram's final losses are from epoch 100, not from the submitted epoch 61 checkpoint. All epochs in this report are 1-based; Shriram's checkpoint file names are 0-based (ckpt_epoch060.pth is epoch 61).

Run histories

Tejas (task3_gan/tejas/results.md, task3_gan/tejas/exploratory/*/metrics_report.csv, task3_gan/tejas/outputs/checkpoint_selection.csv):

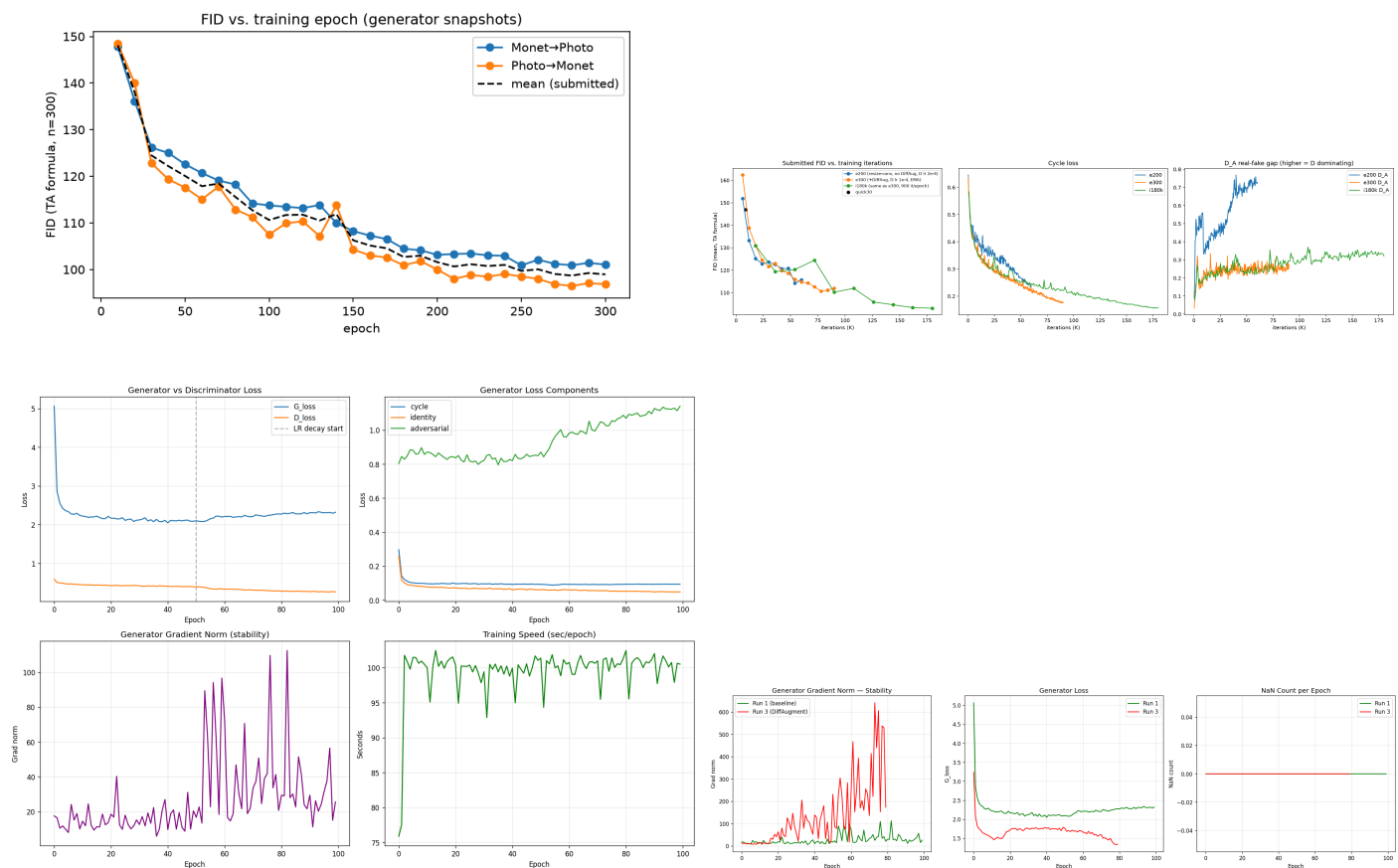
Run	Iterations	Change and reason	Best FID mean (A2B / B2A)	Train time	G grad-norm max	Score
quick30	9K (30 x 300)	Baseline: paper configuration, transposed-conv upsampling	146.97 (142.82 / 151.12)	23.5 min	142.7	-73.70
e200	60K (200 x 300)	Resize-convolution (quick30 showed checkerboard artifacts), longer schedule, snapshot selection	114.23 (122.30 / 106.16)	102.8 min	916.1	-57.33
e300	90K (300 x 300)	DiffAugment, slower D (1e-4) and EMA, because the D dominated in e200 (D loss about 0.06, real/fake 0.86/0.14)	110.70 (115.56 / 105.85)	168.1 min	179.6	-55.56
i180k	180K (200 x 900)	Same setup, twice as long, because the e300 plateau was the LR reaching 0	103.09 (104.70 / 101.48)	329.5 min	190.4	-51.75
i270k (final)	270K (300 x 900)	Direction-specific identity ramp (Photo to Monet had plateaued near 101), D LR 7.5e-5, 45/55 schedule	98.70 (100.92 / 96.49)	507.0 min	1,017.9	-49.55

The score column is $-(\text{FID} + \text{MiFID})/2$. Only i270k has a recorded leaderboard score; the values for the earlier runs come from Tejas's snapshot tables.

Shriram (task3_gan/shriram_dundigalla/results.md, failure_analysis.md, full_metrics_report.csv, src/task3_cyclegan_unet.ipynb):

Run	Change	FID mean (A2B / B2A)	G grad-norm mean / max	NaN
1 (submitted)	Baseline U-Net, λ_{cycle} 10, λ_{id} 5, 100 epochs x 1,000 steps; best checkpoint epoch 61	101.44 (103.92 / 98.95)	27.0 / 112.5	0
2	λ_{cycle} 8, λ_{id} 2, 80 epochs, constant LR for 40 epochs then linear decay; best checkpoint epoch 80	101.51 (102.65 / 100.37)	not measured	0
3	Run-2 losses plus DiffAugment (colour, translation, cutout) on all D inputs, 80 epochs	112.97 (108.54 / 117.40)	137.0 / 640.5	0

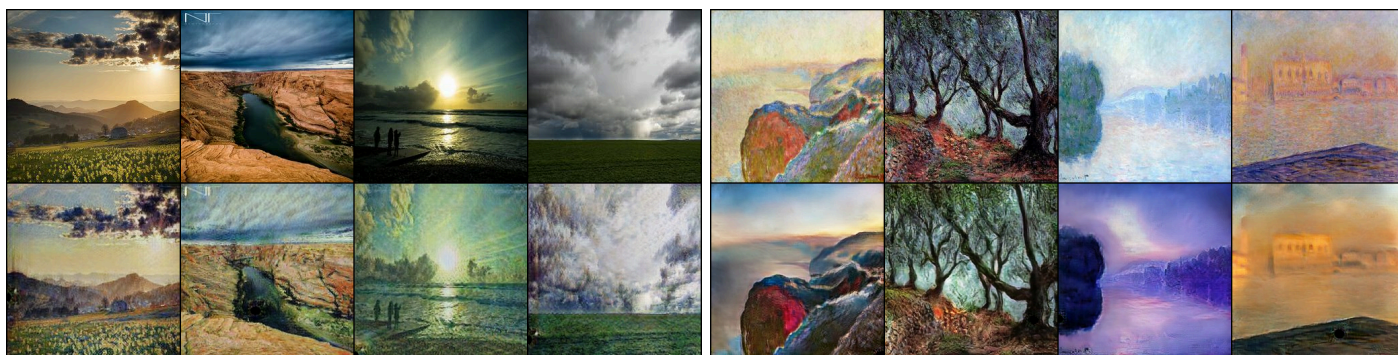
Shriram's run-2 FID is reported from the official script (101.51); his own scoring of the same checkpoint gave 99.58. For run 1 he selected among checkpoints at epochs 51, 61, 71, 81, 91 and 100 with an earlier A2B-only FID (88.02, 87.32, 88.13, 87.59, 87.51, 89.02), which is on a different scale from the course script.



Top: Tejas's FID per snapshot in the final run, and best FID across his five runs. Bottom: Shriram's run-1 losses and gradient norm, and his run 1 vs. run 3 stability comparison.

Example translations

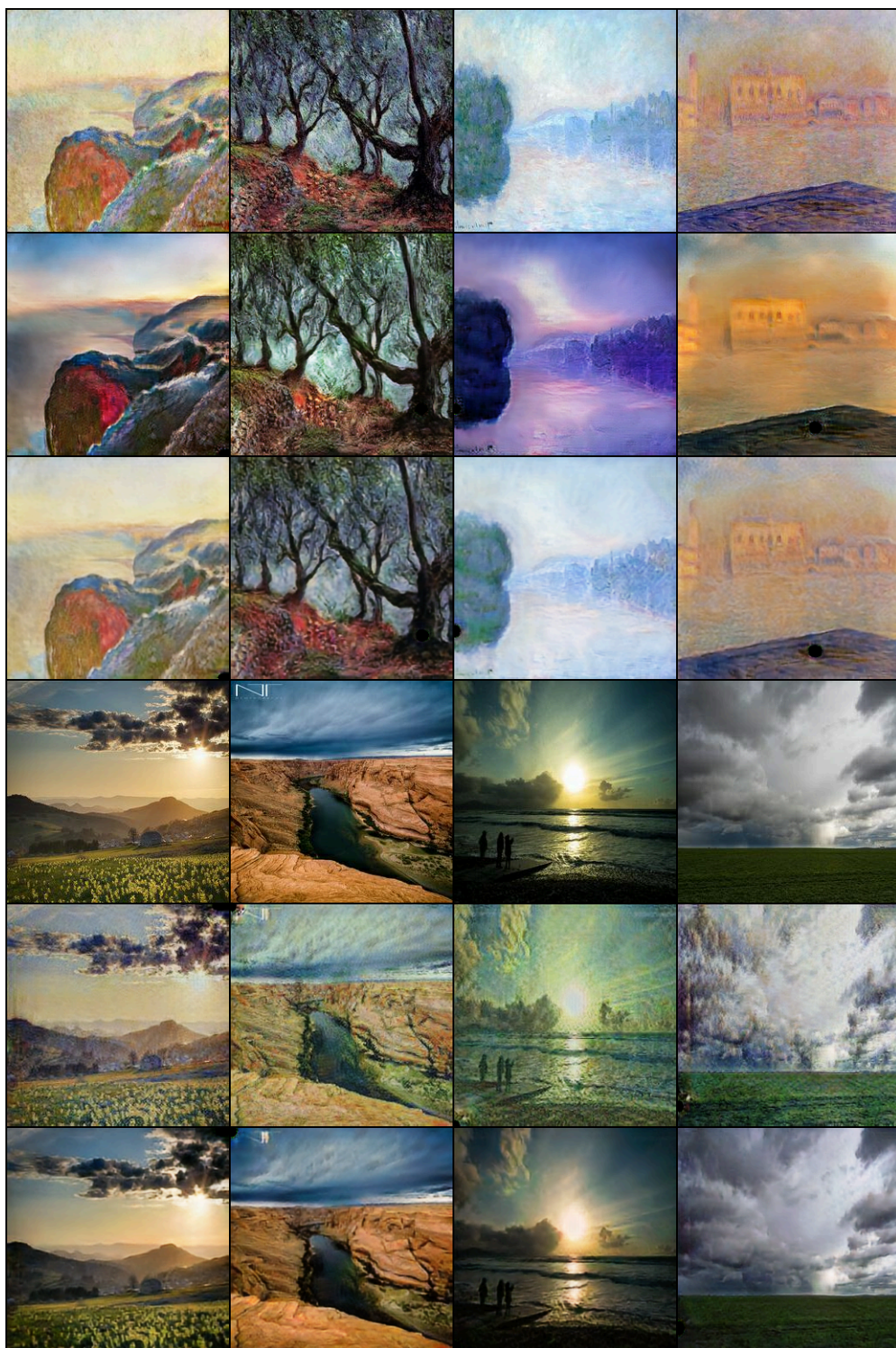
Tejas (i270k, EMA snapshot epoch 280), input and output pairs. Left: Photo to Monet. Right: Monet to Photo.



Shriram (run 1, epoch 61), unedited generator outputs from outputs/pred_B2A/ and outputs/pred_A2B/. Top row: Photo to Monet. Bottom row: Monet to Photo.



Cycle-consistency check. Tejas's $A \rightarrow B \rightarrow A$ reconstructions are shown below (cycle L1 0.0785 / 0.0849). Shriram checked cycle consistency numerically only (cycle L1 0.0471 / 0.0493).



Joint analysis

ResNet vs. U-Net. Tejas's ResNet-9 (28.3M parameters) has lower FID than Shriram's U-Net (114.3M) in both directions: 100.92 vs. 103.92 for A2B and 96.49 vs. 98.95 for B2A. It also changes the input more (LPIPS 0.322 / 0.364 vs. 0.181 / 0.250).

The U-Net reconstructs better through the cycle (L1 0.047 / 0.049 vs. 0.078 / 0.085), which fits Shriram's view that skip connections make reconstruction easy but resist the repainting that stylisation needs. The comparison

is confounded with training length (270K vs. 100K updates), DiffAugment and EMA (Tejas only) and the discriminator LR.

DiffAugment. It hurt Shriram's U-Net (FID 101.51 in run 2 to 112.97 in run 3, G gradient-norm mean 137 in run 3 vs. 27 in run 1). His discriminator was not dominating (D loss about 0.26).

In Tejas's e200 run the discriminator was dominating (D loss about 0.06), and adding DiffAugment with a slower D and EMA improved FID from 114.23 to 110.70. Both results fit the rule in TEAM_PROTOCOL.md: use DiffAugment only when the discriminator overfits. Tejas changed three things at once, so the share due to DiffAugment alone is not isolated.

Training length. Tejas's FID kept improving with longer schedules (110.70 at 90K, 103.09 at 180K, 98.70 at 270K). Shriram's best checkpoint was epoch 61 of 100, so the last 39 epochs did not improve FID even though cycle loss kept falling. For both of us, periodic FID on checkpoints was a better selection signal than training loss.

Direction. Both models score better on Photo to Monet. Both show higher precision and lower recall for A2B (Tejas 0.747 / 0.483, Shriram 0.683 / 0.443) and the reverse for B2A (0.543 / 0.720 and 0.500 / 0.730). B2A density is the weakest value for both (0.451 and 0.385). Tejas's identity ramp, which lowers λ_{id} for Photo to Monet only, moved his B2A FID from about 101 to 96.49.

Stability. No run of either member produced NaNs. Tejas's final run has his largest G gradient-norm spike (1,017.9 vs. 190.4 for i180k) with a similar mean (29.5). The raw log places it at step 2,300, which is epoch 3 at 900 iterations per epoch, early in the constant-LR phase. Training continued normally, so it was a transient and not a divergence.

He also observed slow discriminator creep late in training (real/fake 0.62/0.38 to 0.70/0.30). Shriram's run 1 was stable at G about 2.3 and D about 0.26.

Audit vs. FID. Only Tejas's outputs have been audited; Shriram's audit is pending at time of writing. The style ratings agree with FID on direction: Photo to Monet was rated 4.57 vs. 4.10 and has the lower FID. Content ratings are high (4.70, κ 1.00, partly a ceiling effect), which matches the low cycle L1.

Evaluation caveat. Both members score the images they trained on, as the competition requires, and both selected checkpoints using that FID (Tejas among 30 snapshots, Shriram among 6 checkpoints). The reported FIDs are therefore slightly optimistic. The course "MiFID" is a positional cosine distance without a memorisation threshold, so it is not the Kaggle MiFID.

Strengths. Tejas's ResNet-9 has the lower FID in both directions (submitted 98.70, Kaggle score -49.55, team rank 20). Each change in his run history improved FID, from 146.97 for quick30 to 98.70 for i270k. Shriram's U-Net gives the better cycle reconstruction (L1 0.047 / 0.049), and his run 1 was stable. Audited content preservation for Tejas's outputs is high (4.70, κ 1.00).

Weaknesses. DiffAugment hurt Shriram's U-Net (FID 101.51 to 112.97). The U-Net also changes the input less (LPIPS 0.181 / 0.250), which fits the view that skip connections resist repainting. Both models are weaker

on Monet to Photo, and B2A density is the lowest value for both (0.451 and 0.385). Artifacts appear in 55% of Tejas's audited samples, and his final run had a large transient gradient spike (1,017.9).

Limitations. Only 300 Monet images; one seed per run; Shriram's outputs are not yet audited and his training peak memory was not logged; different GPUs.

Next steps. Finish the shared blinded audit with both members' outputs, hold out photos to measure generalisation, try R1 or spectral-norm regularisation (Tejas), try a weaker DiffAugment policy under a matched budget (Shriram), and compare ResNet and U-Net generators under the same budget.

Evidence

	Tejas	Shriram
FID per checkpoint	fid_vs_epoch.png , checkpoint_selection.csv , experiments_comparison.png	printed in task3_cyclegan_unet.ipynb (runs 1, 2 and 3)
Loss curves	loss_curves.png , epoch_history.csv	loss_curves_run1.png , run1_vs_run3_stability.png , metrics_per_epoch.csv
Examples	examples_photo2monet.png , examples_monet2photo.png , samples/	pred_A2B/ (300), pred_B2A/ (7,038)
Cycle verification	cycle_verification.png	numeric check in the notebook
Submission	submission.csv (98.7030 / 0.41056)	submission.csv (101.4378 / 0.41722)
Course-script cross-check	ta_eval_script_executed.ipynb , submission_ta_script.csv (98.70296 / 0.41056, agrees with submission.csv to $< 1e-6$)	course script run in the notebook with only paths changed: Photo to Monet FID 98.952, Monet to Photo FID 103.924, matching submission.csv
Audit	audit_results.csv , audit_rater1.csv , audit_rater2.csv , audit/	human_audit/ (pending)
Raw logs	t3_cyclegan_tejas_20261002-132424.jsonl and one per exploratory run	task3_cyclegan_unet_run1.log , train.log
Weights	checkpoints/G_AB.pt, G_BA.pt (EMA, committed)	ckpt_epoch060.pth (1.3 GB, not in git; SHA-256 in checkpoints/README.md)

Failure and error analysis

Task 1: three failure cases per member

Tejas (task1_llm/tejas/failure_analysis.md; 300 new characters per prompt):

Case	Decoding / prompt	Snippet (verbatim)	Failure type	Observation
1	Greedy, "One day, Tom" (repeated 4-gram rate 0.213)	"One day, Tom wanted to play with his toys and run around outside. He saw a big tree with many leaves. Tom wanted to climb the tree, but he was too small. He tried to climb the tree, but he could not. Tom was sad. Tom saw a big tree..."	Repetition loop	The plot stalls and restarts the same scene; argmax reinforces frequent phrases.
2	Greedy, "The little dog"	"The little dog was so happy to have a new friend. They played together all day long. The end. / Once upon a time, there was a little girl named Lily. She loved to play outside in the sunshine. One day, she saw a big box in the garden."	Story reset	After "The end." the model falls back to the most common opening, identical to the greedy "Once upon a time" output. The 300-character generation is longer than the 256-character context.
3	T = 0.8, "In the forest"	"In the forest was a bright, sunny barn. The barn was full of lots of animals, flowers and flowers. At night, the barn went for many long walks in the meadow. But then, something bad happened. A man came to say he had a spell."	Semantic error	Grammatical but impossible: a barn "went for many long walks". Good local syntax, weak world knowledge.

Shriram (task1_llm/shriram_dundigalla/failure_analysis.md; prompt "Once upon a time", 400 new characters):

Case	Temperature	Snippet (verbatim)	Failure type	Observation
1	0.0 (greedy)	"...She saw a big box of stars and a big box. She wanted to see what was inside. She wanted to see what was inside. She took a big bite and started to cry."	Verbatim repetition	Repeated 4-gram rate 0.060 at T = 0 vs. 0.000 at T ≥ 0.5 in this draw; a decoding failure.
2	0.8	"One day, Timmy wanted to go to the park, so he asked his mommy if he could have him playing. His mommy bought the nuts and said hello. Timmy remembered his mommy tost his toys and she asked him what he was wrong."	Broken grammar and a non-word	"tost" is not a word; distinct-n still scores the passage as perfect, which led him to add a training-vocabulary word rate (0.988).
3	1.0	"Once upon a time, there was a little boy named Benny. Benny loved his car and rolled his doll's teeth. ... They were talking to the park to play them with it. / Once upon a time, there was a little boy named Timmy."	Loss of coherence, early story boundary	Benny leaves the 128-character window and the model starts a new story mid-generation.

Both members found the same two mechanisms: greedy looping, and story resets caused by the context window and the story separator. This supports both proposed fixes, a repetition penalty (already shown to work by Shriram) and a longer context.

Task 2: 20-error reviews

Both members reviewed 5 confident false positives, 5 confident false negatives, 5 near-threshold errors and 5 weakest-slice errors.

Tejas, BiGRU with attention, weakest slice truncated = True (task2_sentiment/tejas/failure_analysis.md, outputs/error_candidates.csv):

Error type	Count
Truncation / long narrative	5
Contrast, mixed, lukewarm or comparative	6
Sarcasm / irony	2
Edited reviews (temporal mismatch)	2
Label noise / rating and text mismatch	2
Domain vocabulary, implicit sentiment, tokenization	3

Examples:

- **FN4** ($p = 0.001$, sarcasm): "TERRIBLE SERVICE, RUDE WAITERS WITH A PISS POOR ATTITUDE! WOULD EAT HERE AGAIN! A++++". Lowercasing and punctuation removal remove the cues.
- **FP5** ($p = 0.999$, comparative): "Though I'm a Copper enthusiast... Maharani was a cheaper but tasty option... Copper is definitely still my place, but Maharani was fine enough." The praise is for a competitor.
- **NT3** ($p = 0.498$, contrast and tokenization): "...deserve 5 starshowever the procedures used to estimate and charge... are in need of improvement." The missing space merges two informative words into one unknown token.

Proposed fix (Tejas). Raise max_len from 188 to 400 tokens (about the 99th percentile), keep head and tail truncation, and retrain Exp 2 with the same seed. Measure the error rate and macro-F1 on the truncated slice before and after, run a paired McNemar test on the full test set, and record training time and peak memory.

Shriram, TextCNN, 2,492 errors (6.56%), weakest slice exclamations = no_exclaim (task2_sentiment/shriram_dundigalla/failure_analysis.md, outputs/error_review_exp2_textcnn.csv):

Error type	Count
Negation scope / detached negation	3
Temporal contrast (past negative, present positive)	2
Genuinely mixed sentiment (near threshold)	5
Probable label noise	2
Late verdict reversal / concessive	2
Sarcasm	1
Target confusion	1
Truncation	1
Rhetorical question	1
Insufficient evidence (very short)	2

Examples:

- **A1** ($p = 0.9994$, sarcasm): "Yay. Cafeteria food. Yay."
- **A2** ($p = 0.9991$, late verdict reversal): "We had a great time and I thoroughly enjoyed the food, but I need to give this a 1 start given that if we had eaten a week later, this salmonella outbreak could have caused immeasurable harm to our child."
- **B3** ($p = 0.0006$, target confusion): "Someone deleted my review.. intentionally. That is very rude and disrespectful. Don't be silly guys.. I like this place and new owner is very nice and friendly..."

Proposed fixes (Shriram). His first priority is position-aware pooling: add a max-pool over the last 20% of tokens to the global max-pool, and check that accuracy improves on "but"/"however" reviews while overall accuracy holds. He also proposes negation-scope marking (a NOT_ prefix on the 3 tokens after a negation) and hand-labelling the 200 most confident errors to estimate the label-noise ceiling.

Both reviews conclude that part of the remaining error cannot be fixed by modelling. Shriram counts 8 of 20 (sarcasm, label noise, mixed reviews). Tejas counts 4 of 20 as unfixable (edited reviews, label noise) and 2 as hard (sarcasm).

Task 3: failure cases

Tejas (task3_gan/tejas/failure_analysis.md):

Case	Failure type	Evidence file	What changed or would fix it
quick30	Checkerboard texture and dark blobs (FID 146.97)	exploratory/quick30/outputs/examples_monet2photo.png	Resize-convolution in e200 (FID 114.23)
e200	Discriminator domination (D loss 0.057 / 0.061, D_A real/fake 0.863 / 0.134 at epoch 200)	exploratory/e200/outputs/epoch_history.csv	DiffAugment, D LR 1e-4 and EMA in e300 (FID 110.70)
e300	FID plateau (110.70 to 111.99) caused by the LR reaching zero	exploratory/e300/outputs/checkpoint_selection.csv	Schedule twice as long in i180k (FID 103.09)
i180k	Photo to Monet stuck near 101 while Monet to Photo improved	exploratory/i180k/outputs/checkpoint_selection.csv	Direction-specific identity ramp in i270k (B2A 96.49)
i270k	Sky speckle and washed-out sun (artifacts in 55% of audited samples); slow D_A creep	outputs/examples_photo2monet.png, outputs/audit_results.csv	R1 or spectral norm, lower identity weight (proposed, not tested)

Shriram (task3_gan/shriram_dundigalla/failure_analysis.md):

Case	Failure type	Evidence file	What changed or would fix it
Run 3	DiffAugment made FID worse (101.44 to 112.97; G gradient-norm mean 27.0 to 137.0)	full_metrics_report.csv	Use DiffAugment only when the discriminator overfits; run 1 D loss stayed near 0.26
Run 2	Loss-weight tuning changed almost nothing (101.44 to 101.51)	full_metrics_report.csv	Bottleneck is likely the architecture or the amount of Monet data, not the lambda values
Run 1	Best checkpoint at epoch 61, not 100, while cycle loss kept falling	logs/metrics_per_epoch.csv	Select checkpoints with periodic FID, not training loss

Case	Failure type	Evidence file	What changed or would fix it
Run 1	Photo to Monet density 0.3847 (weakest value); skip connections resist repainting	full_metrics_repo rt.csv	Points to a limitation of the U-Net choice

Both members found that DiffAugment helps only when the discriminator is dominating: it helped Tejas's e200 setup and hurt Shriram's balanced run. Both also found Photo to Monet harder to fix: Tejas's i180k run stalled in that direction, and Shriram's weakest value (density 0.3847) is in that direction.

Reproducibility

Smoke tests. README.md gives a one-command smoke test for Tejas's notebooks: `cd task3_gan/tejas/src & SMOKE=1 jupyter nbconvert --to notebook --execute --output smoke_run.ipynb task3_cyclegan.ipynb`, with the same pattern for Tasks 1 and 2.

Shriram's scripts take a `--smoke` flag that writes only to gitignored smoke/ folders: `python task1_llm/shriram_dundigalla/src/train.py --smoke --require-cuda` and `python task2_sentiment/shriram_dundigalla/src/train.py --smoke --require-cuda`. His Task 3 notebook has its own one-epoch smoke cell. Repository-wide checks are python scripts/verify_gpu_ready.py and python scripts/verify_submission.py.

Configuration. Tejas uses a CFG cell in each notebook, recorded in each manifest JSON. Shriram uses YAML configs (task1_llm/shriram_dundigalla/src/config*.yaml, task2_sentiment/shriram_dundigalla/src/config.yaml, task3_gan/shriram_dundigalla/src/config.yaml).

Raw logs and manifests.

Member	Raw logs	Manifests
Tejas	reproducibility/raw_logs/{task1_llm,task2_sentiment,task3_gan}/tejas/ (Task 3: final run and four exploratory runs)	reproducibility/manifests/{task1_llm,task2_sentiment,task3_gan}/tejas/ (JSON and requirements_frozen.txt)
Shriram	reproducibility/raw_logs/ (task1_gpt_char_*.log, task2_*.log, task3_cyclegan_unet_run1.log, smoke logs)	reproducibility/manifests/shriram_dundigalla_manifest.md

Data and weights. The datasets were provided by the instructor and are not stored in git. They can be downloaded from [Google Drive](#); README.md gives the expected folder layout. Shriram's Task 3 checkpoint (1.3 GB) is not in git; its SHA-256 is in task3_gan/shriram_dundigalla/checkpoints/README.md.

Absolute paths in logs and manifests. Raw logs and manifests are kept unedited, so they contain the runtime paths of the machines that produced them (a Windows PC and Google Colab). This is disclosed in README.md. Tejas's notebook outputs use <REPO> placeholders. Shriram's Task 3 notebook keeps its Colab Drive paths, so running it elsewhere requires changing the Drive root in its first cell.

References

1. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. *NeurIPS 2017*.
2. Eldan, R., & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? *arXiv:2305.07759*.
3. Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. *ICCV 2017*.
4. Zhao, S., Liu, Z., Lin, J., Zhu, J.-Y., & Han, S. (2020). Differentiable Augmentation for Data-Efficient GAN Training. *NeurIPS 2020*.
5. Joulin, A., Grave, E., Bojanowski, P., & Mikolov, T. (2017). Bag of Tricks for Efficient Text Classification. *EACL 2017*.
6. Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification. *NeurIPS 2015*.
7. Johnson, J., Alahi, A., & Fei-Fei, L. (2016). Perceptual Losses for Real-Time Style Transfer and Super-Resolution. *ECCV 2016*.
8. Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *ICML 2017*.
9. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. *NeurIPS 2017*.
10. Bińkowski, M., Sutherland, D. J., Arbel, M., & Gretton, A. (2018). Demystifying MMD GANs. *ICLR 2018*.
11. Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. *CVPR 2018*.
12. Naeem, M. F., Oh, S. J., Uh, Y., Choi, Y., & Yoo, J. (2020). Reliable Fidelity and Diversity Metrics for Generative Models. *ICML 2020*.
13. Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-Image Translation with Conditional Adversarial Networks. *CVPR 2017*.
14. Mao, X., Li, Q., Xie, H., Lau, R. Y. K., Wang, Z., & Smolley, S. P. (2017). Least Squares Generative Adversarial Networks. *ICCV 2017*.
15. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI 2015*.
16. Odena, A., Dumoulin, V., & Olah, C. (2016). Deconvolution and Checkerboard Artifacts. *Distill*.
17. Holtzman, A., Buys, J., Du, L., Forbes, M., & Choi, Y. (2020). The Curious Case of Neural Text Degeneration. *ICLR 2020*.
18. Keskar, N. S., McCann, B., Varshney, L. R., Xiong, C., & Socher, R. (2019). CTRL: A Conditional Transformer Language Model for Controllable Generation. *arXiv:1909.05858*.
19. Kim, Y. (2014). Convolutional Neural Networks for Sentence Classification. *EMNLP 2014*.
20. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8).
21. Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. *EMNLP 2014*.